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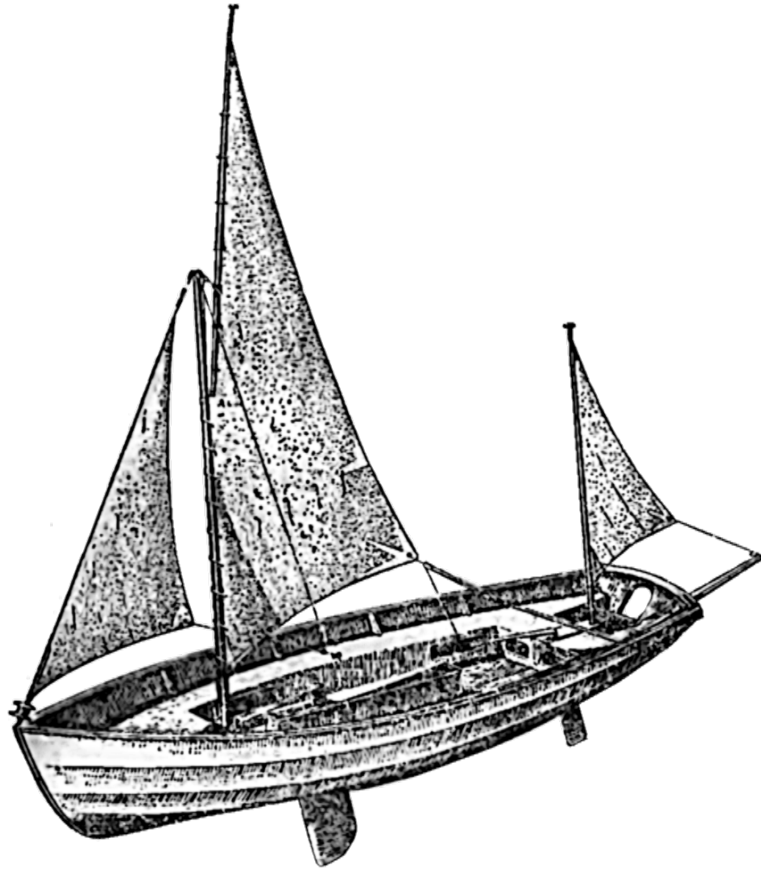
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Be advised : the information, diagrams and plans within this site are downloadable free-of-charge, and are offered in good faith without any warranty or guarantee. All known information is included here 'as is' - no further support with regard to the building of this vessel is offered, or will be made available.

Site last updated: 29 Aug 2011

1 PROJECT PALTUS - AN INTRODUCTION ...

Project Paltus is a website devoted to the 'Paltus' boat design by the Russian designer Dmitry Kurbatov. This site contains copies of the original published texts, and translations of these into English, for which I'd like to thank Petr from Siberia for his help with corrections and clarifications. The responsibility for accuracy, of course, remains my own.

There is also a page devoted to a comparison between Kurbatov's design and his recommended building methods and those of the Drascombe Lugger, upon which his design was based.

Although Kurbatov and his 'Paltus' design are well-known in Russia, almost nothing about them is known in the West, and it is hoped that these webpages will encourage those who wish to build a proven sea-worthy boat of this size and type to consider Kurbatov's Paltus. These pages may also provide a possible solution for those wishing to build a Drascombe Lugger in wood, now that the plans for these are no longer available.

1.1 THE LUGGER "PALTUS"

When the Russian designer Dmitry A. Kurbatov (1936-1993) saw the sales brochures of (the original) Honnor Marine, builders of the Drascombe series of boats, he immediately saw the potential of the Drascombe Lugger and set about drawing-up a set of construction plans for a similar craft, which he called the "Paltus", which in English translates to either Halibut or Turbot. A strange choice of name, for both of these are flat fish !

The first and most complete published version of the 'Paltus' plans were published in the magazine "Power and Sail Boats" (Katera I Yahty) No.72 in 1978, under Kurbatov's pen name D. Antonov. Later, the 'Paltus' drawings were included in the third edition of "15 Boat Projects for Amateur Building", a book published in Leningrad, 1985, so these plans may be readily sourced elsewhere, albeit with the accompanying text in Russian. I have included copies of these original texts and diagrams in Appendices A and B.

Of Kurbatov it has been said, "The multifaceted talent of this renaissance man permitted it to not only develop and realize many interesting projects, but to become a unique observer of the most important foreign events, the instigator of new trends in the area of small and sport vessel construction. Dmitry Kurbatov was, in the literal sense of the word, a legend of domestic amateur boat building."

Within his plans, Kurbatov recommends that the hull be constructed 'the right way up' using temporary construction frames which are later removed to be replaced by the final transverse frames. In contrast, builders of wooden Drascombe boats in England have preferred to construct their hulls 'upside-down', (i.e. keel uppermost), with the final transverse frames also acting as construction frames.

Details of both methods are included within these pages, and it will be left to the individual builder to choose whichever method he (or she) considers best.

1.2 BUILDING KURBATOV'S 'PALTUS'

Translated from the Russian

The attention of amateur boatbuilders is drawn to the proposed project, which has been developed from the sales materials of Honnor Marine, builder of the Drascombe series of boats. By careful study of this project, a real possibility exists for the home builder, using readily available materials.

The hull is sheathed using strips of 7mm thick waterproof plywood ¹, using the 'edge-upon-edge' or clinker method. Plywood is also used in the construction of frames, onboard seats and deck. During construction, the strakes are glued together and rivetted with 3mm copper nails/rivets and copper washers at 60-80mm intervals. The double-thickness clinker overlap (which should be within the limits of 15-20 mm) serve as longitudinal stiffening stringers which makes possible the construction of a strong and rigid hull with the additional use of only five frames. Even further strength and rigidity are provided by the vertical walls and horizontal surfaces of seating and decks.

Another, more contemporary method of building the hull with the same "angular" contours is proposed: the strakes may be joined by clips made from copper wire, with epoxy resin and glass tape being applied to both sides of the join. The copper clips only provide a convenient method of assembly, the ultimate strength of the join being dependant upon the epoxy and glass fibre tape. One Leningrad boatbuilder has even employed synthetic thread in place of the copper wire in the construction of an 8-metre plywood yacht.

Both methods of construction are approximately equivalent in terms of labour, expense and reliability. However, many supporters of clinker construction consider that their method deflects spray and helps to moderate rolling in heavy sea. One criticism made of the 'stitch and tape' method, however, is that if the gluing of the fibreglass tape is inadequate, moisture penetration with rapid rotting of the wood may follow. Thus it is necessary to ensure that the seams are covered adequately on both sides with glass tape (*using several layers - see later*) and epoxy glue.

Assembly of the hull may be more conveniently performed on the construction mold in the normal position, i.e., with the keel downwards. The strakes are then applied to the previously assembled and carefully positioned transverse frames, keel, stem and transom. When constructing the mould, the dimensions of frames 2, 4, 6 and 8 (as taken from the table of offsets) must be adjusted to take into account the thickness of the external plywood skin, as the measurements given refer to the external hull surface. Within the table, measurements of the chines are considered to be that of the upper edge of the strake.

1 The '7mm waterproof plywood' to which Kurbatov refers is a special grade available in Russia. If using WBP or marine plywood, then the next size up should be used.

Table of Offsets	Frame Number											
	0	1	3	3	4	5	6	7	8	9	10	Tr.
	Height from the Base-Line, mm											
Keel	260	61	10	0	7	25	51	88	128	173	221	-
Strake 1	355	300	250	215	186	176	179	190	210	235	263	265
Strake 2	520	456	405	360	325	310	310	324	348	378	405	415
Strake 3	740	680	624	580	550	530	522	534	560	593	628	667
Sheer	965	910	864	825	792	778	762	760	760	770	780	-
	Half-widths from centre-line, mm											
Sheer	286	510	689	820	900	945	955	932	880	795	685	545
Strake 3	230	466	655	800	895	945	955	932	880	765	685	545
Strake 2	140	363	551	699	795	848	860	838	784	705	608	555
Strake 1	52	225	380	510	605	660	671	649	595	514	415	-

The stem and keel must be faired before attaching them to the building mould: the stem-piece should be bevelled to accommodate the bunching together of the strakes where they will join the stem. The required angle can be determined either by a lath or a length of the plywood used for the strakes.

Before cutting a strake from a plywood sheet, a full-length template should be made from low-quality plywood or hardboard. After attaching the template to the curved mould by panel-pins, the lines of the strake are extended outwards from the mould and marks made on the inside of the template. After removing the template, these marks may be joined by using a flexible lath to indicate the position of the next strake. (With the exception of the sheer-strake, an allowance of 15-20mm should be added for overlap. Then place this template onto the high-quality plywood to be used for the strake, and transfer the line by pricking-through with the point of an awl, before finally cutting-out the strake.

The first strake to be applied is the lower bottom, and is attached to the keel by 4mm x 30mm wood screws, spaced at 60 mm intervals. At the stern, the next strake will present at an abrupt angle which makes the overlap unsuitable for riveting. Here it is better to simply butt its edge against the previous strake and secure with copper wire clips and epoxy and glassfibre tape as described above, after trimming to ensure that the strake fits snugly against the bottom strake. The adjoining edge surfaces are thoroughly cleaned, and coated with glue before rivetting.

The longitudinal clinker strakes must become flush at the stem. Thus for a distance of approximately 800 mm it is necessary to chamfer the outer edge of the strake and the lower inner edge of the next strake of the clinker join.

The strakes are temporarily attached to the mould by nails, so that when the entire hull is riveted, the moulds may be taken out and replaced by frames, with wood screws being employed to secure the skin to the frames, using the nail holes as a guide. Rather than fairing each frame so that the planking will have flat surfaces to glue and fasten to, installation of the frames may be simplified by bevelling only those frame components which make contact with the inner surface of hull (i.e. the frame battens) to provide a flat contact surface - then the plywood frame itself may be attached to those previously bevelled parts, directly into place.²

2 Taking 'Frame - Station 2' as an example (see Frames diagram), cut the plywood frame with a flat (90 degree) edge. Then cut wooden battens 58 and 51, and bevel them to fit the hull curvature, before attaching them to both the hull and plywood frame with screws and glue. Then cut-out and attach batten 60. The frame is now completely installed into the hull.

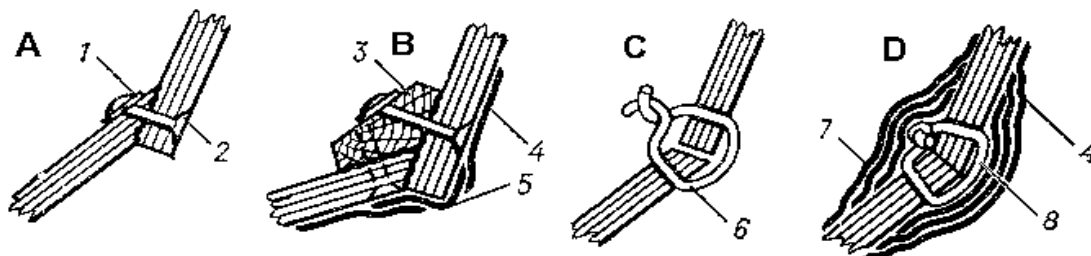
To further simplify this process, the hull can be pulled gently away from the mould using suitable lengths of wood. It is necessary to have previously made cut-outs in the frames for the longitudinal chines of the strakes.

The described boat uses a heavy centreboard, cut from steel plate of thickness 12 - 14 mm. It's weight is about 45 kg which, when added to the dynamic loads of landing on a sand-bar or even simply when rolling, requires a centreboard case of substantial strength. The base, knees and supporting timbers of the centreboard case should be made from oak, with the walls of the case being made from 7 - 10mm exterior plywood, the case being secured to the keel by bolts, with waterproof glue being used throughout. A second similar well, in a smaller size, is provided for a lifting rudder. Before assembly, a layer of diluted glue should be applied to the internal surfaces of both wells, as a sealing coat.

In spite of the high water resistance of exterior plywood, it's open edges require protection against the penetration of moisture. Two or three layers of fibreglass tape with a width of 25-40mm should be laid on all outer edges, with a wider overlapping layer superimposed upon these. Covering the joins using just one piece is not an option, as voids may be created along the angles, into which water will subsequently penetrate.

Methods of joining hull strakes:

A = 'edge-upon-edge' (clinker) B = using a batten (at the transom)
C = using a copper clip D = clip & fibreglass tape (finished)



1 - copper washer; 2 - copper rivet; 3 - batten; 4 - fibreglass tape (main)
5 - extra fibreglass tape at edge; 6 - copper wire clip Ø 2-2.5mm
7 - three layers of fibreglass tape over butt join; 8 - buried copper clip

For rowing, the boat must be fitted with a pair of removeable transverse thwarts (with the sizes 28 x 250 x 1200 mm), which would be located above the longitudinal bank-seats and prevented from moving along the boat by simple bosses or lugs attached to boards at their leading and trailing edges. The optimum length of oars for this boat is 3300 mm. (10'10")

The unsinkability of the boat may be assured by providing hermetic sealing of compartments beneath the longitudinal bank-seats (for example between stations 4 and 6), although it would be necessary to provide hermetically-sealed inspection holes in the vertical walls of the compartments for their periodic inspection, repair and drying. Alternatively, the area beneath the bank-seats could be filled with closed-cell foam buoyancy.

Anticipating questions from readers as to what opportunities exist for the substitution of waterproof plywood by other materials, we shall answer as follows:

for the building of this boat it is possible to use a sheet glass-fiber-reinforced plastic, fiberglass laminate and even the more usual construction-grade plywood, providing that several layers of

fiberglass fabric are applied on top of the skin in order to achieve the desired rigidity and durability. A thickness of GRP skin of 4-5 mm would be sufficient, especially if reinforced by ribs made from plastic foam. Even if these materials are not available, it would still be possible to build the hull from glued strip-plank laths, or even fir or pine boards, glued edge to edge.

The main-mast, mizzen and gunter spars are of continuous, round cross-section, however it is preferred that these be made from two or even four pine laths, as glued spars are more durable and there is less chance of them warping with fluctuations of humidity.

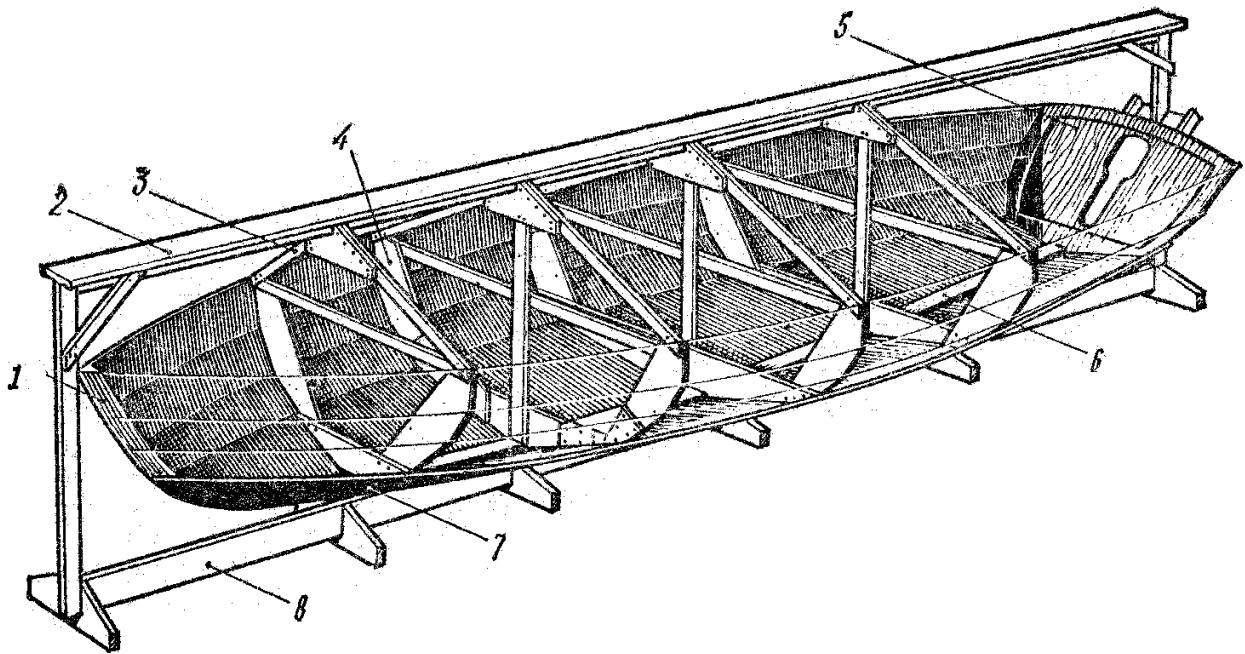
Regarding the sails and rigging of the boat: the mizzen-mast has no standing rigging - it stands in a metal mast step, and is supported by wooden partners at the top of the rudder well. The mainmast is supported by a pair of shrouds, and a forestay - which can be provided by the steel luff-wire 'bolt-rope' of the foresail if jib roller-furling is fitted. This roller-furling consists of a drum (6), attached to the tack of the jib or genoa, being freely rotating relative to it's deck fitting (10). The head of the sail is attached to the halyard by means of a swivel (2), which ensures free rotation of the foresail's luff-wire or bolt-rope without twisting the halyard. Thin synthetic rope (7) is wound on the drum, so that when it is unwound the foresail is reduced to a dense roll.

The running end of this control rope (7) is led back into the cockpit for ease of access by the helmsman.

Thus, at anchor, or at the approach to land, it is not necessary to lower the sail from it's mast. To restore the sail to it's working state, all that is required is to release the control rope, and unfurl the sail by pulling on it's sheets. Of course it is still possible to attach the foresail in the usual way, using piston-hanks or clips hooked onto the forestay.

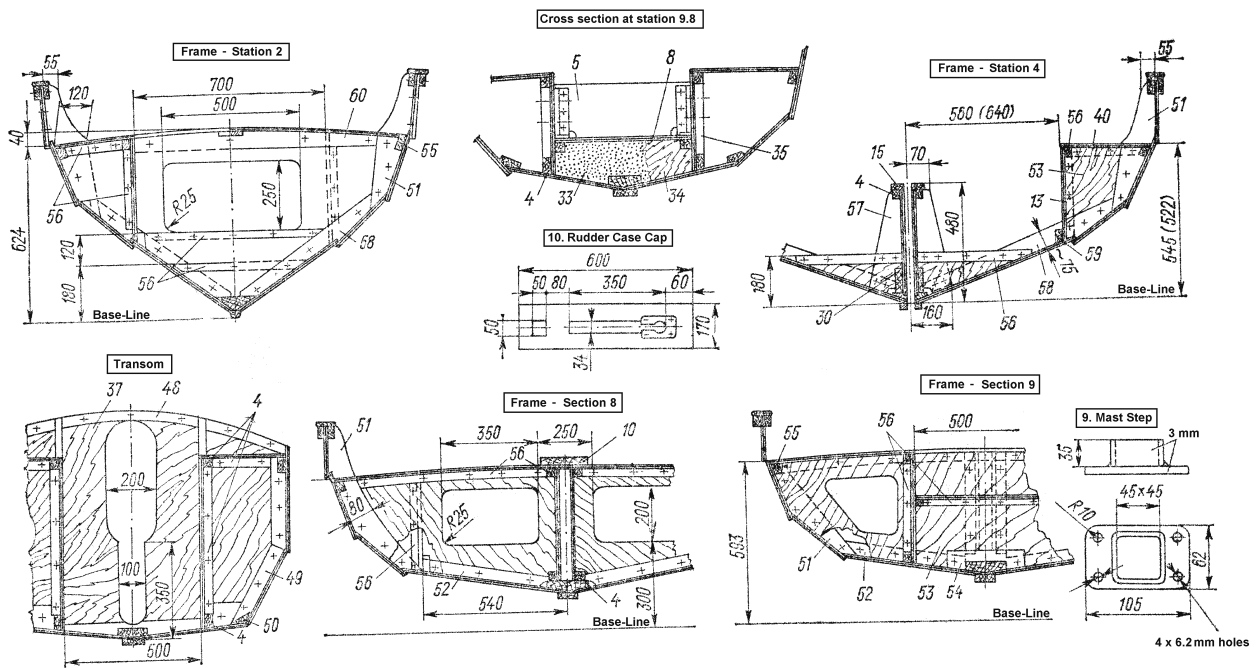
The mizzensail-boomkin must be made detachable; for it will simplify mooring by the stern to a high quay and simplify management of the boat when sails are not being used.

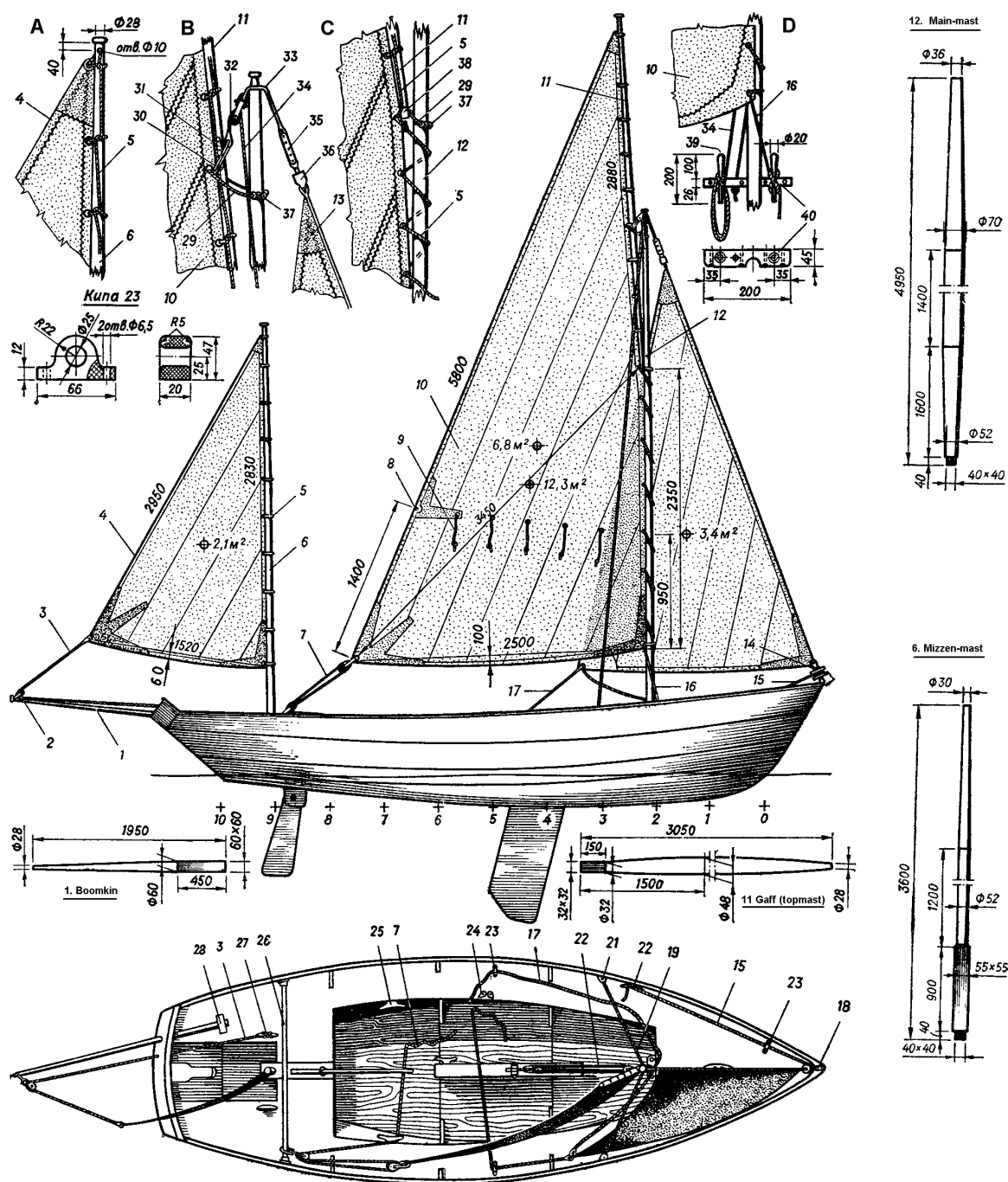
Installation of the main-sheet of the "Paltus" may be the same as on the original "Drascombe Lugger", when it is possible to control the sail by means of one sheet, and during tacking it is not necessary to adjust the sheet with each turn - the sail automatically passing from side to side, although it is recommended that the sheet's cleat be positioned close to the centreline so that on both tacks the sheet length is approximately equal.



Pic 343. Moulds and jointed stem, keel and transom on the building stock.

1 - stem, 2 - upper strongback, 3 - support of the mould, 4 - mould, 5 - transom,
6 - keel, 7 - garboard strake (bottom strake), 8 - lower strongback.





344. The Sails and Rigging of the Boat.

A: - mizzen-masthead; B: - mainmast masthead;
C: - attachment of mainsail and gunter to the mainmast;
D: - attachment of mainsail halyard and mainsail tack - rope.

1 - mizzensail-boomkin, 2 - mizzensail sheet block, 3 - mizzensail sheet, 4 - mizzensail 2.1 sq.m. ,
5 - furling line, synthetic rope $\phi 4-6 \text{ mm}$, 6 - mizzen mast, 7 - main sail sheet, 8 - reef grommet,
9 - reef becket, 10 - mainsail - 6.8 sq.m. , 11 - upper mast, 12 - main mast, 13 - jib 3.4 sq.m. ,
14 - jib furling drum, 15 - jib furling control rope, 16 - main sail tack-rope, 17 - jib sheet,
18 - stemhead, 19 - wire shroud, $2.5 - 3.2 \text{ mm}$, 20 - jib furling cleat, 21 - shroud chainplate,
22 - centreboard tackle, 23 - jib sheet slide, 24 - jib sheet cam cleat, 25 - main sheet cleat,
26 - main sheet traveller, 27 - mizzen sheet cleat, 28 - mizzensail boomkin step, oak,
29 - parrel, synthetic rope $\phi 6 \text{ mm}$, 30 - upper mast furling line, synthetic rope $\phi 6 \text{ mm}$,
31 - cheeks, oak, 32 - mainsail halyard block, 33 - strap, 34 - mainsail halyard,
35 - forestay adjustment plate, 36 - Snap shackle with swivel, 37 - parrel beads, plastic,
38 - gaff bail bracket, 39 - belaying pin.

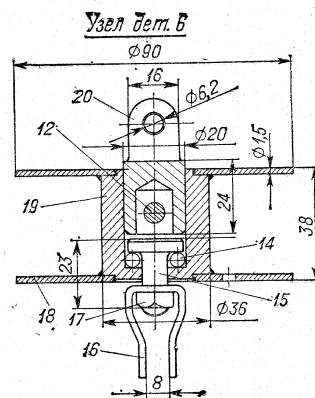
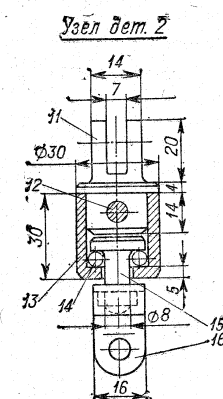
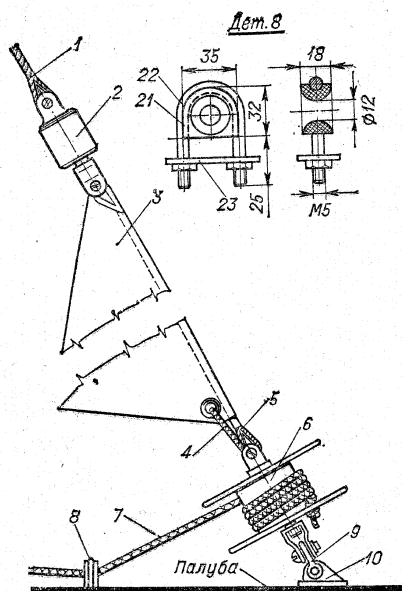
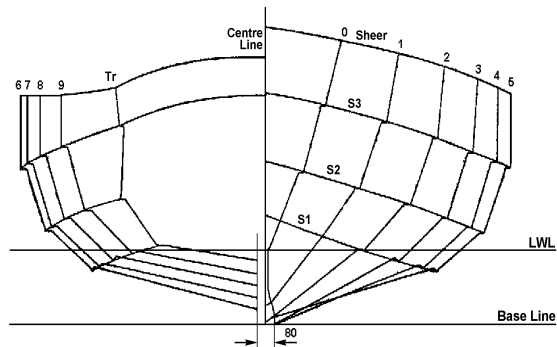
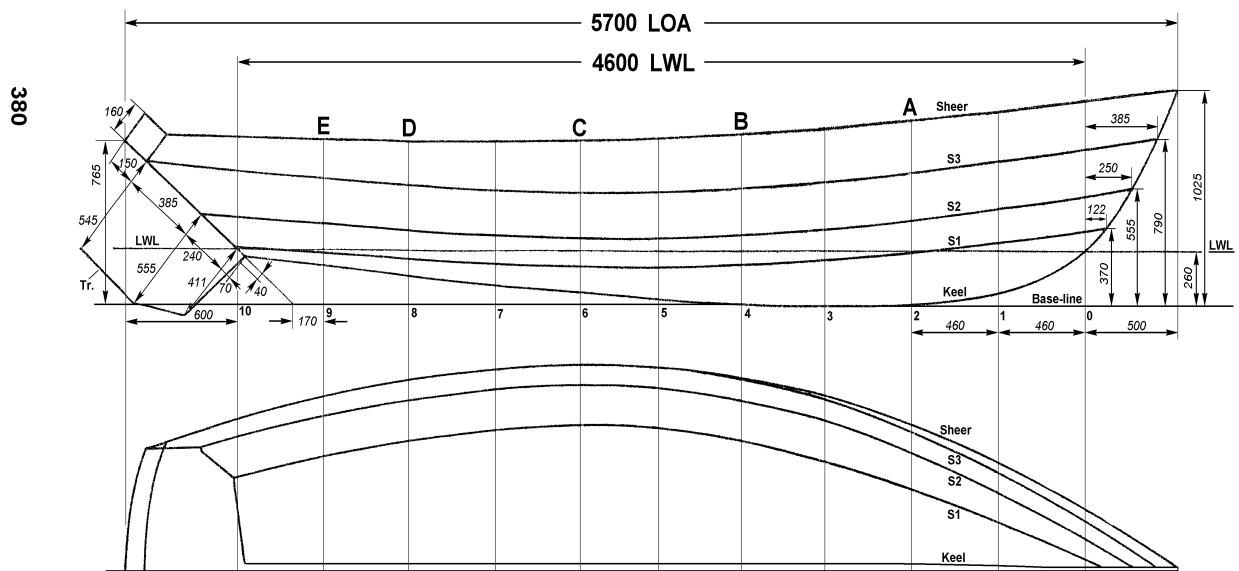


Рис. 345. Конструкция устройства для закрутки стакселя.

1 — стаксель-фал; 2 — верхний вертлюг; 3 — парус; 4 — галс-оттяжка; 5 — коуш ликтроса; 6 — барабан; 7 — шнур капроновый; 8 — киповая планка; 9 — скоба; 10 — обушок; 11 — вилка; 12 — штифт $\varnothing 6$; 13 — корпус вертлюга; 14 — шарик $\varnothing 3-4$; 15 — болт; 16 — скоба, $\delta = 2$; 17 — гайка М8 (закернить); 18 — щека барабана; 19 — корпус барабана (сплав АМг); 20 — вилка; 21 — колодка (текстолит); 22 — скоба $\varnothing 5$; 23 — планка 2×15 .

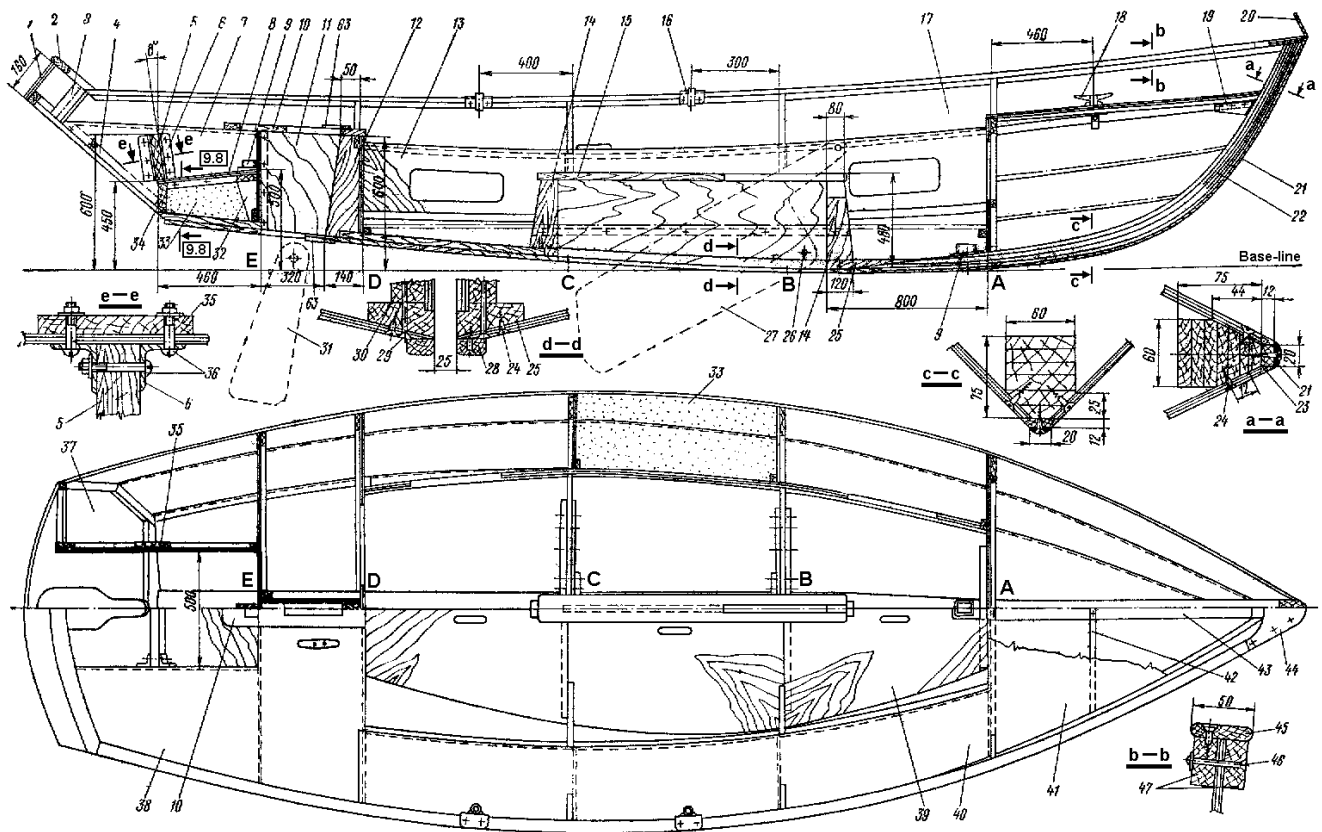
1.3 LINES AND OFFSETS



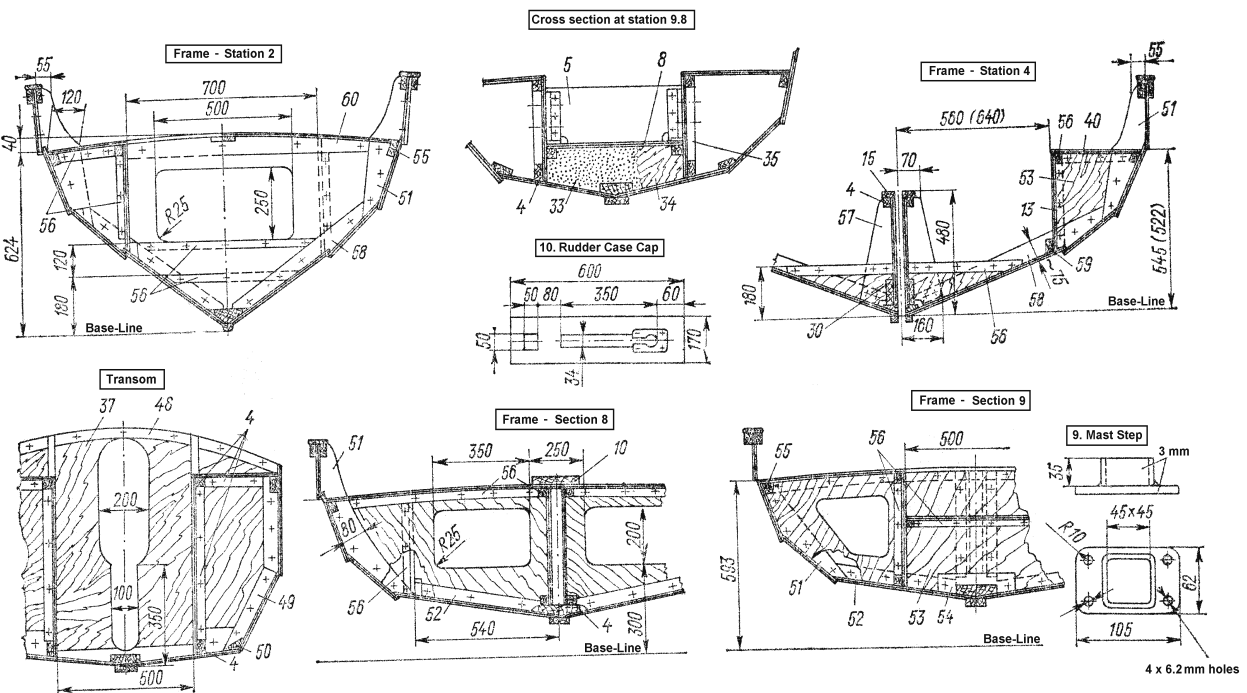
Lines of the Luger "ПАЛТУС" ("Paltrus" or Turbot)

Table of Offsets	Frame Number										
	0	1	②	3	④	5	⑥	7	⑧	⑨	10 Tr.
Heights from base-line, mm.											
Keel	260	61	10	0	7	25	51	88	128	173	221 —
Strake 1	355	300	250	215	186	176	179	190	210	235	263 265
Strake 2	520	456	405	360	325	310	310	324	348	378	405 415
Strake 3	740	680	624	580	550	530	522	534	560	593	628 667
Sheer	965	910	864	825	792	778	762	760	760	770	780 —
Half-widths from centre-line, mm.											
Sheer	286	510	689	820	900	945	955	932	880	795	685 545
Strake 3	230	466	655	800	895	945	955	932	880	795	685 545
Strake 2	140	363	551	699	795	848	860	838	784	705	608 555
Strake 1	52	225	380	510	605	660	671	649	595	514	415 —

1.4 SECTIONS AND PARTS



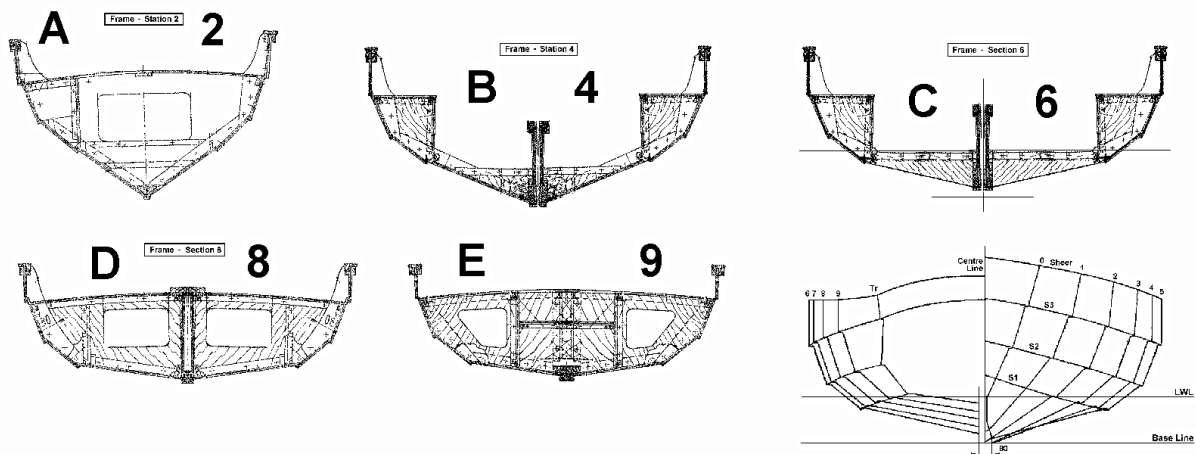
1. taffrail, 2 layers of 6mm plywood. 2. taffrail cap, 20 x 50, oak. 3. angled beam, 40 x 80. 4. wooden lath, 28 x 28. 5. outboard engine board, 150 x 150 x 38
6. angle iron, 32 x 32 x 3, L=150. 7. outboard well longitudinal, 7mm plywood. 8. outboard pan, 7mm plywood. 9. metal mast step. 10. rudder case cap, 7mm plywood
11. rudder case wall, 7mm plywood. 12. rudder well key, 34mm, oak. 13. open day locker, 5-6mm plywood. 14. centreboard well key, 25mm, oak. 15. centreboard case cap, 22 x 110, oak. 16. rowlocks. 17. skin, sides and bottom, 7mm plywood. 18. mooring cleat, 220mm. 19. deck support knee. 20. forestay tang 22mm x 4mm, steel
21. stem band, s/steel 16mm x 4mm D section, fastened with woodscrews 16mm x 3.5mm, spaced 100mm. 22. stem piece, 8 laths 60mm x 10mm glued together and chamfered
23. outer cover plate, oak, 35mm x 12mm. 24. C/S woodscrew 30mm x 4mm, staggered at 40mm separation (29 off). 25. keel/centre-board support, 150mm x 30mm
26. centreboard pivot bolt 14mm dia. 27. centreboard, 12-14mm, steel, 45Kg. 28. false keel (wear strips) 80mm x 20mm, oak. 29. C/S woodscrew 6mm (24 off)
30. centreboard case support 100mm x 28mm. 31. rudder blade, 4-6mm, steel. 32. support, 20 x 100. 33. buoyancy foam. 34. lower engine board, 500mm x 220mm x 30mm
35. engine board support, 120mm x 20mm. 36. bolt, 5mm. 37. transom, plywood 7mm. 38. quarter deck, plywood 6mm. 39. bottom (floor)boards, plywood, 6mm
40. side decks, 6mm plywood. 41. foredeck, 6mm plywood. 42. stringer, 25 x 60. 43. foredeck support, 22 x 100. 44. stem cover plate, 12mm plywood
45. rubbing strake cap, 50mm x 12mm, oak. 46. copper rivet, 3mm. 47. rubbing strake, inner & outer, 40mm x 20mm. 48. transom beam, 25 x 40 (three laths glued together)
49. transom binding battens, 25 x 60. 50. chine support stringer, 40 x 25, L=500. 51. framing, 22mm. 52. framing, 22 x 40. 53. bulkhead, 6mm plywood
54. inner keel, 22 x 80 x 300. 55. outer side-deck support stringers, 22 x 40. 56. lath, 22 x 30. 57. centreboard case support knee, 22 x 160, oak. 58. framing, 22 x 75
59. lath, 22 x 30. 60. deck support beam, 22 x 70. 61. rudder head, pipe, 30mm x 3 mm, steel. 62. rudder head cheek, 3mm x 130mm x 170mm. 63. rudder head bearing & cover plate, brass



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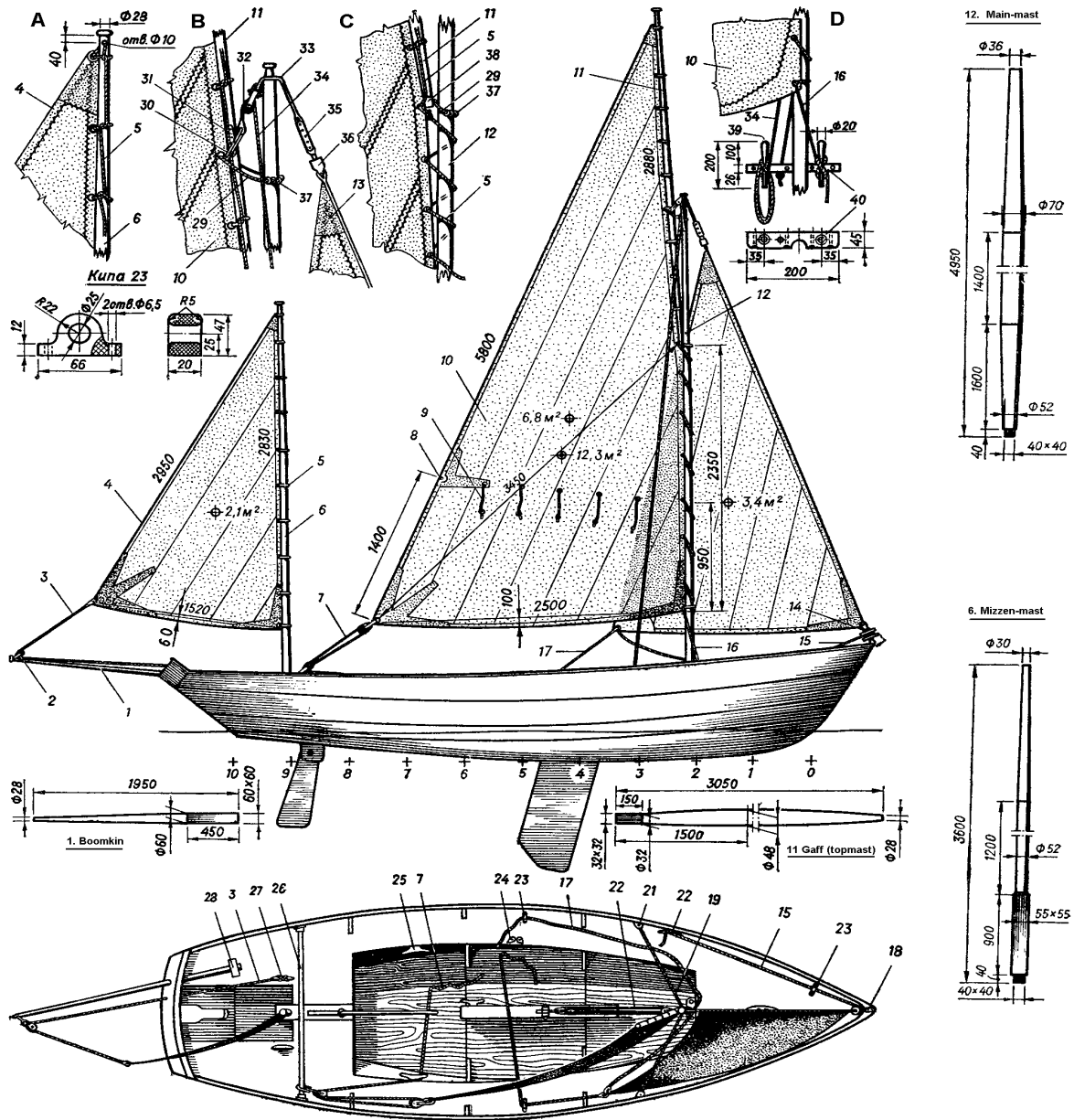
1.6 FRAME CHECK

The construction frames have been labelled A, B, C, D and E, which correspond to stations 2, 4, 6, 8 and 9. In this diagram I have generated a frame for station 6, in order that all the construction frames may be superimposed on the lines cross-section diagram (which is included here at the same scale) by using suitable graphics software, in order that the accuracy of the frames may be confirmed. They appear accurate.





1.7 SPARS, SAILS & RIGGING

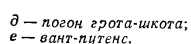
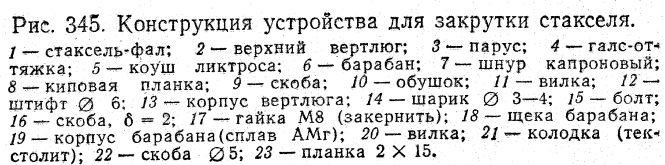


344. The Sails and Rigging of the Boat.

- A: - mizzen-masthead; B: - mainmast masthead;
 C: - attachment of mainsail and gunter to the mainmast;
 D: - attachment of mainsail halyard and mainsail tack - rope.

1 - mizzensail-boomkin, 2 - mizzensail sheet block, 3 - mizzensail sheet, 4 - mizzensail 2.1 sq.m., 5 - furling line, synthetic rope $\varnothing$ 4-6mm, 6 - mizzen mast, 7 - main sail sheet, 8 - reef grommet, 9 - reef becket, 10 - mainsail - 6.8 sq.m., 11 - upper mast, 12 - main mast, 13 - jib 3.4 sq.m., 14 - jib furling drum, 15 - jib furling control rope, 16 - main sail tack-rope, 17 - jib sheet, 18 - stemhead, 19 - wire shroud, 2.5 - 3.2 mm, 20 - jib furling cleat, 21 - shroud chainplate, 22 - centreboard tackle, 23 - jib sheet slide, 24 - jib sheet cam cleat, 25 - main sheet cleat, 26 - main sheet traveller, 27 - mizzen sheet cleat, 28 - mizzensail boomkin step, oak, 29 - parrel, synthetic rope $\varnothing$ 6mm, 30 - upper mast furling line, synthetic rope $\varnothing$ 6mm, 31 - cheeks, oak, 32 - mainsail halyard block, 33 - strap, 34 - mainsail halyard, 35 - forestay adjustment plate, 36 - Snap shackle with swivel, 37 - parrel beads, plastic, 38 - gaff ball bracket, 39 - belaying pin.

(not translated)



2 THE DRASCOMBE[®] LUGGER

Translated from the Russian

These open boats can be found in large numbers in almost in any sailing centre around the coast of England and on large lakes and rivers inside the country. Quite often "Drascombes" can be found at sea far from the coast, where they cope well on steep waves, maneuvering under their characteristic low-aspect sails. The firm "McNulty Boats" have produced five models of glass-fibre "Drascombe" boat, differing only slightly from the original "Drascombe Lugger" (1964). The secret of the success and popularity of a boat of this type can be found in a design which ensures a high level of seaworthiness, an attractive appearance, high quality manufacturing, and unpretentiousness in operation: in simplicity of the control of sails and efficiency when navigating under low power engines.

LOA, m	5.7
LWL, m	4.6
Beam max, m	1.91
Beam, laden waterline, m	1.52
Draft, board up, m	0.26
Draft, board down, m	1.2
Dry weight with rigging, kg	380
Waterline displacement, kg	550
Sail Area, m ²	12.3
Power of outboard motor, HP.	3-8
Speed with a 5HP motor, km/h	11

2.1 THE DRASCOMBE[®] LUGGER

This boat was developed in 1964 by the former owner of the Totnes shipyard and former Royal Navy officer John Watkinson, when he decided to leave small shipbuilding and study agriculture at his farm on Dartmoor. But John could not completely say goodbye to the sea and soon he began to think about building a boat for himself and his family for day trips and fishing. The future vessel had to be capacious - large enough to carry the whole Watkinson family - stable, seaworthy, easy to control, and to have comparatively low weight so that it would be possible to transport it on a trailer behind a family-sized car. Furthermore, John wanted the boat to be capable of giving an experienced sailor a lively and exciting sail, and to develop sufficient speed with the outboard motor to overcome the flow of the river Dart.

In 1965 the "Drascombe Lugger", as the designer named his creation, was successfully launched in the river Dart and it successfully underwent further trials in the estuary (the mouth of which is subject to strong seas and tidal currents) and in the adjacent coastal waters of Southern England. As Watkinson did not intend to enter into mass production, the hull was of wooden construction, the lines of which were inspired by the coble working boats of England's North-East coast, which themselves can trace an ancestry back to the Vikings.

The boat that John hand-built in a barn on his farm at Drascombe Barton was an immediate success and its obvious commercial potential prompted him to initiate production of the boats in GRP. Other models followed, but all followed the original philosophy of safety, robustness, and fun.

The hull of the first boat had a length of 5.72 m with a beam of 1.9 m and formed from strips of 9 mm waterproof plywood, with 4 planks on each side, so in appearance it resembled the clinker construction of the original working boats. Inside the perimeter of the hull were fixed wide longitudinal bank-seats, imparting extra rigidity in combination with several frames, which are also made from water-resistant plywood. These bank-seats proved to be very convenient for the counterbalancing of the boat when sailing, with crew leaning back against the support of the bulwarks.

The designer has provided opening-scuppers in the boards at seat level for the draining of water. If the "Lugger" should ship a wave or should it become necessary to put the boat on an even keel after capsizing, water above the seats will drain outboard, with the bulwarks remaining above water level. Spaces under the bank-seats were filled with blocks of foam plastic, which ensured unsinkability. This cockpit arrangement provides sufficient accommodation for 4-6 people on a short trip, together with the necessary equipment for a more prolonged journey.

The boat was fitted with a heavy centre-board, cut from 13 mm steel sheet. It weighed 55 kg, and when lowered increased the draft to 1.22 m, noticeably increasing the stability of the vessel. The rudder blade was made from 4 mm steel plate, welded to the rudder head and, as with the centre-board, it descended into its own well. In shoal waters it was possible to lift the rudder and steer the boat with the aid of a steering oar, for which a special recess was provided in the transom. The draft with the plates lifted did not exceed 0.26 m.

A third well was made in the transom for an outboard engine. The engine sits within that housing and is thus protected from wave damage from over the stern. In the case of a breakdown at sea it is possible to repair the engine without hanging out over the transom., and when sailing the engine can be tilted back with its propellor clear of the water, so avoiding drag.

The boat was equipped with a triangular foresail, mizzensail and mainsail with a total area of 12.26 m². The solid section masts were made from glued laths: the mizzen-mast having no standing rigging (the area of the mizzensail being only 2.04 m²), but the mainmast is supported by a forestay and a pair of shrouds. This "one-and-a-half- mast" arrangement is very convenient for trolling for fish, as under just foresail (3.34 m²) and mizzensail the boat can head into the wind at low speed, or stably drift. The mizzensail also proves to be useful during navigation under engine, or when lying at anchor when it holds the bow into the wind and waves, thus decreasing the rolling motion.

The sails were boom-less, simplifying their control by unskilled crew. It is no longer necessary to fear being struck on the head by the boom during tacking. Likewise, gybing is less dramatic. With a boom, the sail moves sharply across the boat with its dynamic force being transferred to the mast and rigging, a process which is sometimes accompanied by an overturning of the boat, but the boom-less mainsail will move from board to board without this dynamic force. This is familiar to every yachtsman who has had to change a mainsail or trysail in a storm.

The luffs of the main and mizzensail are laced to, and may be kept furled around their masts. The foresail was also fitted with a roller-furling mechanism to roll the sail around the forestay. In order to get under way using sails, it is sufficient to release the canvas on both masts, and the vessel is ready. Likewise, not more than three minutes are required to douse the sails - it is even possible to remove the masts with the sails still attached to them, which can then lie within the boat. The mizzensail is sheeted via a boomkin.

The Drascombe Lugger has proved to be an ideal family boat - a day sailer for navigation in the rough coastal waters around England. It was very stable, and in response to sudden squalls the "Lugger" heeled only to the level of the scuppers - heeling did not increase even during further strengthening of the wind - thanks to the wide beam, heavy centreboard, and low sail area. With a 6HP outboard engine the boat developed a speed of 5.5 knots (10 km/hr), although being light it is also easy to row, especially with one person on each oar.

2.2 BUILDING A WOODEN DRASCOMBE LUGGER:

The hull is built from 9mm marine plywood. The frames, bulkheads, centreboard case and rudder trunk are made from 12mm marine plywood. It is recommended that Iroko hardwood be used for floors, gunwales, frame doublers and stem laminates. The decks are made from 6mm marine plywood, and the masts and spars are made from Columbian Pine or Douglas Fir, as is the inwale, to achieve a fair curve at deck level.

The forward bulkhead, midship frames, aft bulkhead, transom, centreplate case, rudder trunk and outboard motor well are all pre-cut and reinforced with hardwood or marine plywood doublers at various points, and assembled prior to fastening them to the building jig.

The rudder trunk is glued and fastened to the aft bulkhead and transom along with the outboard motor well, and the centreplate case is glued and fastened to the midship frames. The components are then fastened to the building jig using temporary fastenings. The hull is built upside down on the building jig.

The frames are then tied together with an inwale at deck level and a hog and inner stem laminates. The fair-up of the frames then takes place and the wide garboard plank is fitted.

The next stage is to plane the plank land to create a joint surface for the next plank, it is vitally important that the joints are accurate, because the hull has very few fastening in it when finished, and you cannot edge-set a plywood plank as you would a normal timber plank.

The next plank when dry fitted is then pre-drilled to take small diameter bolts at approx. 6 inch spacing, these acting as a temporary clamping system, fastening the planks together until the glued joints have cured. The whole of the boat is of glued construction using phenolic resin glues such as Aerodux 500 or Cascophen. The bolts are removed at a later stage. The procedure is similar for the next plank.

The hull is now built to deck level and a general fair up takes place, prior to fitting the keelson and the outer stem laminates. The outer stem laminates are glued together using 3inch x ½ inch coach screws with wooden pads to spread the load; they are screwed through the pre-drilled laminates into the inner stem laminates that were fitted before any planking was added. When the glue has fully cured, the bolts and coach screws are removed and hardwood dowels glued in their holes to give a complete solid wood construction for all of the joints and very few fastenings in the hull at all.

The build has now reached the point at which the outside of the hull can be faired up and an almost complete finish achieved, even though the boat is only built to deck level with just 3 planks. The hull can now be released from the building jig and turned the right way up. The process to completion can begin with a clean up of the inside of the hull, cutting off all of the hardwood dowels which make the boat look like an inverted porcupine, cleaning up any glue excess and fitting the deck beam, carlands and ancillary reinforcing blocks.

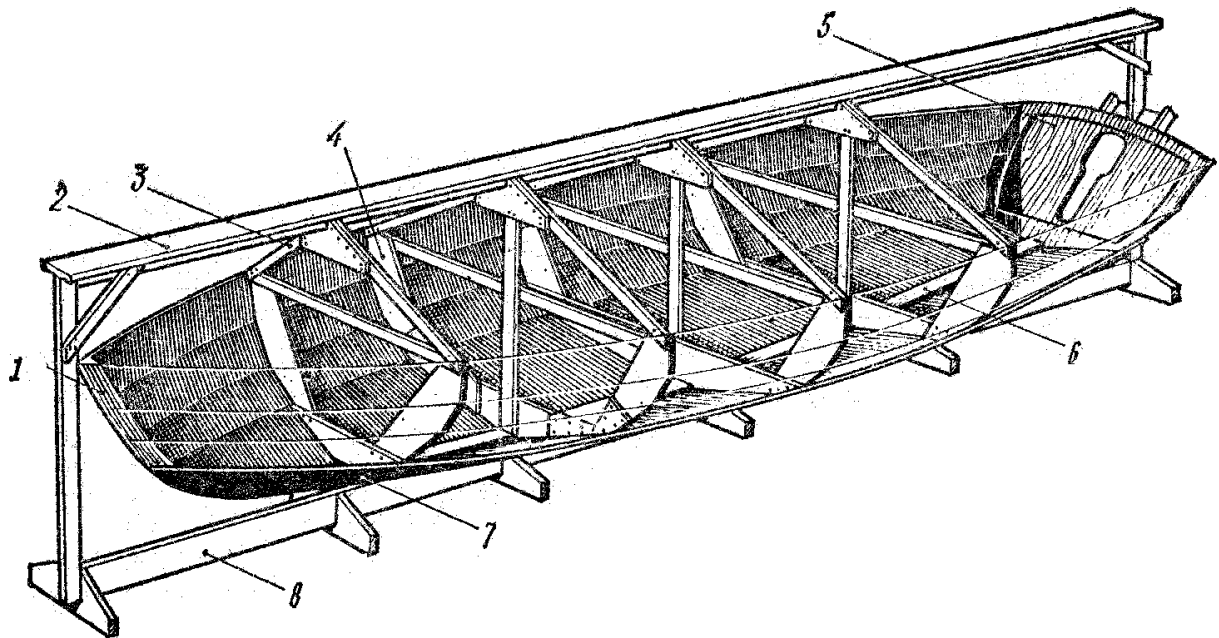
The next stage is to dry fit the decks. When done, they are removed to allow painting of the bilges up to the deck level and also painting the underside of the decking, this makes for ease of working prior to the decks being glued and fastened down. When the deck is complete, the plank land for the top strake can be faired up, the knees are then fitted and the top strake followed by the transom return, the quarter knees, breasthook and the laminated hardwood gunwales.

The rest of the woodwork is visible wood work and particular attention has to be paid to the detail, the decks are covered in a 16oz woven roving glass fibre cloth, giving extra stiffness to the 6mm ply decking and a reasonably non slip finish. The masts and spars are made from Columbian Pine (Douglas Fir), with a set of masts and spars comprising main mast, mizzen mast, yard and bumkin. The "Drascombe Longboat" is essentially a stretched "Lugger", being built on the same jig with a 3ft centre section added.

3 COMPARISON

Perhaps the first difference to note between the Paltus and Drascombe Lugger is the preferred building method:

Method used for the Paltus



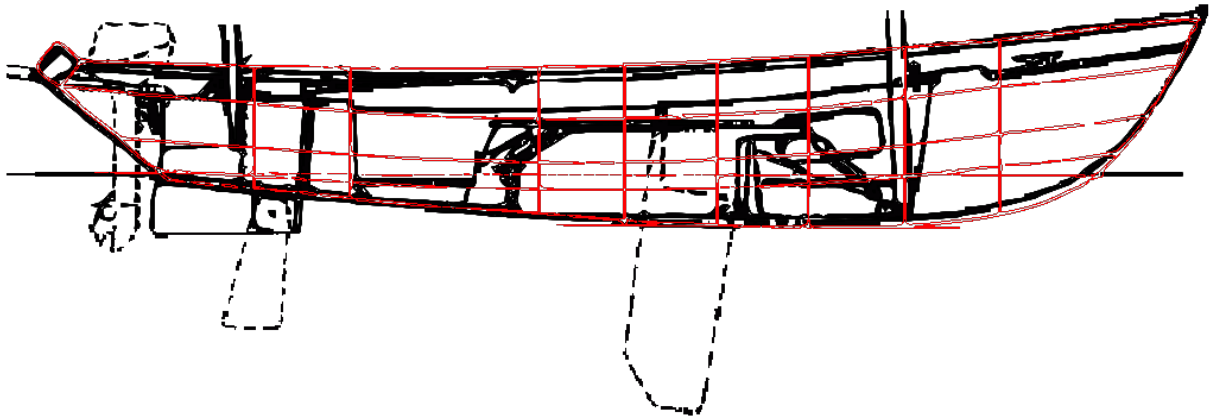
Pic 343. Moulds and jointed stem, keel and transom on the building stock.

1 - stem, 2 - upper strongback, 3 - support of the mould, 4 - mould, 5 - transom, 6 - keel, 7 - garboard strake (bottom strake), 8 - lower strongback.

Method used for the Drascombe Lugger



The Paltus is a design developed directly from the Drascombe Lugger - Kurbatov has said this - and although they certainly appear similar at first sight, they are not absolutely identical. Although their plan views are very similar, in profile at least 3 small differences can be seen:- the transom angle is slightly different to that of the Drascombe Lugger; there is slightly less sheer rocker; and the forefoot is slightly deeper.



An examination of the boat's interior reveals rather more significant differences: perhaps the most obvious being the sealed buoyancy tanks fitted to the Paltus, which are similar in concept to those of the Mirror dinghy. Less obvious perhaps is the lateral support for the centre-board case: the D/Lugger having a single central support, whereas the Paltus has two - one at each end of the case.

Interior view of a new Paltus build.



Interior view of a Drascombe Lugger renovation.

